

SOP-02 Processing, Quality Control and Packing (Factory)

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Document control

Field	Detail
SOP number	SOP-02
Title	Processing, Quality Control and Packing (Factory)
Version	1.0
Owner	Founder & Managing Director, StoneCrest International
Applies to	Every export order, first container and repeat orders alike
Used by	StoneCrest QC staff, factory production supervisor, factory labour manager
Previous SOP in sequence	SOP-01 Block Selection and Reservation (Quarry)
Review	Every 12 months, or after any quality complaint

Purpose and scope

Every piece of an order is cut, finished, checked and packed to the same standard, from the reserved blocks only, and nothing leaves the factory until it passes final inspection. This SOP covers QC Stages 2 to 6 of StoneCrest's 6-stage QC process; Stage 1 is SOP-01.

Scope: all processing for export orders at every factory StoneCrest produces through: finished monument pieces, slabs and cut-to-size stone. The first container and every repeat order follow the same steps.

The reference for every check is the Approved Reference for the order (see SOP-01) and the retained reference chip from the selected block. Every check measures against these, never against the factory's own standard.

Roles and responsibilities

The factory produces; StoneCrest checks and releases. No piece is packed, and no container is loaded, without the StoneCrest QC Inspector's sign-off.

Role	Responsible for
Order Owner (StoneCrest)	Issues the Order Specification Sheet and cutting plan approval; final release of the container
QC Inspector (StoneCrest)	Runs the checks in Stages 2 to 4 and 6; accepts or rejects pieces; keeps records and photos
Factory Production Supervisor	Plans and runs cutting, calibration, polishing and edge finishing to the specification; stops work on instruction
Factory Labour Manager	Assigns trained operators to each machine; handles pieces with care; separates rejected pieces; carries out packing and loading as instructed

Finished-piece quality standard and tolerances

Every finished piece must meet all of the standards below. These are the same standards stated to buyers, and a piece outside any one of them is rejected. Where an order specification sets a tighter figure, the order specification applies.

Item	Standard
Colour	Uniform deep black, matching the Approved Reference and the reference chip
Grain and tonal variation	Very fine, even grain; minimal tonal variation within the piece and across the order
Veins and marks	No visible white or grey veins, spots, streaks or patterns
Cracks and chips	No cracks, hairlines, chipped corners or broken edges
Thickness	Within ± 1 mm of the specified thickness, checked at all four corners and the centre
Length and width	Within ± 1 mm of the specified size
Squareness	Corners at 90°; diagonals equal within 2 mm
Flatness	Polished face flat, with no visible dips or waves under a straight edge
Polish	Durable mirror polish on all visible sides, even across the face, no dull patches, swirl marks or burns
Edges	Finished smooth, with no sharp edges or chips
Bottom face	Sawn (not polished), unless the order specifies otherwise
Natural stone	No dye, oil, wax or colour enhancer used at any stage

Block receipt and cutting plan

Production starts only on blocks reserved under SOP-01, and only after the cutting plan is approved.

1. **Receive blocks.** Check each block number against the Block Reservation Record. Photograph each block on arrival and note any transit damage. A block without a matching number is not used.
2. **Store separately.** Keep the order's blocks together, labelled with the order number, apart from other stock.
3. **Cutting plan.** The Production Supervisor prepares a plan for each block: which sizes come from which block, and in what quantities. Large pieces are planned first on the clearest faces; smaller pieces use the rest.
4. **Quantity ranges.** Plan the quantity of each size within the buyer's minimum and maximum, aiming for the planned quantity, so that the container can be balanced at packing by priority order.
5. **Approval.** The QC Inspector approves the cutting plan before the first cut. The plan is attached to the order file.

Cutting and calibration, with random production inspection (QC Stage 2)

Pieces are cut and calibrated to size, and checked at random during the run so that any problem in a block is caught before the whole block is cut.

1. **Mark each piece.** As each piece is cut, mark it on the sawn bottom face with the order number, block number and a piece number, in chalk or marker that will not damage the stone.
2. **Cut to size.** Cut each piece to its specified size plus the finishing allowance the factory uses for polishing and edging.
3. **Calibrate thickness.** Calibrate each piece to its specified thickness.
4. **Random inspection.** The QC Inspector pulls at least 1 piece in every 10 from each block, and at least 3 pieces per block, during the run. Check colour against the reference chip, veins, cracks, thickness and size.
5. **If a random check fails:** stop cutting that block. The QC Inspector checks the pieces already cut from it. If the defect runs through the block, the block is withdrawn and replacement volume is arranged under SOP-01.
6. **Record.** Log each random check (piece number, result, photo) on the production log.

Polishing and edge finishing, with in-line review (QC Stage 3)

Polish is checked as it is applied, so that dull patches or burns are corrected on the line, not found at final inspection.

1. **Polish in full sequence.** Polish every visible side through the factory's full abrasive sequence, from coarse to final polish. No grit step is skipped to save time.
2. **No treatments.** No dye, oil, wax, resin or colour enhancer is used on the stone at any stage.
3. **Edges.** Finish all edges smooth, with no sharp edges or chips.
4. **In-line review.** The operator checks each piece under good light as it comes off the polisher. The QC Inspector checks at least 1 piece in every 10 on the line for evenness of polish, swirl marks, burns and edge finish.
5. **Rework on the line.** A piece with a polish defect goes back through polishing. A piece with a chip, crack or colour defect is set aside for the QC Inspector.
6. **Wash and dry.** Wash each finished piece with clean water and dry it before it moves to final inspection.

Final inspection of every piece (QC Stage 4)

Every piece in the order is inspected, not a sample, before it is packed. Only pieces marked "Pass" go into a crate.

1. **Light and position.** Inspect each piece dry, face up, under good daylight or equivalent lighting, beside the reference chip.
2. **Visual check.** Colour, grain, tonal variation, veins, spots, cracks, chips and polish on every visible side, against the standard table in this SOP.
3. **Measure.** Thickness at four corners and the centre; length and width; both diagonals. Record the readings.
4. **Acetone check.** On at least 1 piece in every 20, wipe a polished face with acetone on a clean white cloth. The cloth must stay clean.
5. **Decision.** Mark each piece Pass or Reject on the Final Inspection Checklist, with its piece number. Rejected pieces are moved out of the packing area at once.
6. **Photos.** Photograph a selection of passed pieces from each size, including a tape measure on the thickness, for the buyer's pre-shipment photo set.

Third-party inspection on buyer request (QC Stage 5)

When the buyer chooses third-party inspection, their inspector gets full access and the container waits for the inspector's release.

- **Arrangement:** the buyer appoints the inspector and chooses the standard and report format. The cost is agreed with the buyer before the order.
- **Timing:** the inspection takes place after StoneCrest's final inspection and before packing is closed, so the inspector can see individual pieces.
- **Access:** the inspector can see all pieces, the reference chip, the block records from SOP-01, and the final inspection records.
- **Findings:** any piece the inspector rejects is handled as a non-conformance under this SOP. The container is not loaded until the inspector's report is received.

Packing, loading and pre-shipment photographs (QC Stage 6)

The container is packed to reach the target weight within the buyer's quantity ranges, and it is not sealed until packing, labels and documents are checked and photographed.

1. **Crates.** Pack pieces in wooden crates of heat-treated or fumigated timber, sized to the pieces. Fumigation is certified for every shipment.

2. **Face protection.** Place protective sheets between polished faces so no polished face touches another piece or timber. Pieces stand on edge or lie flat as the crate design requires, with spacers so they cannot move.
3. **Strapping and labels.** Strap every crate. Label each crate with the order number, crate number, sizes, piece count and gross weight.
4. **Balance the load.** Target 27.5 tonnes gross including packing, within ± 500 kg. Keep every size between the buyer's minimum and maximum. If more or less weight is needed, adjust priority 1 first; if priority 1 is at its minimum or maximum, move to priority 2, and so on.
5. **Load and lash.** Load crates with the heaviest at the bottom and weight spread evenly along the container floor. Brace and lash every crate so it cannot shift at sea.
6. **Check before sealing.** The QC Inspector checks crate count, labels, lashing, and that the packing list matches what is loaded.
7. **Photographs.** Photograph packed crates, the container at each stage of loading, the final load before the doors close, and the seal number. Send the photos to the buyer before the container moves.
8. **Packing list.** The final packing list shows actual pieces per size, m³ and gross and net weight. The commercial invoice is based on the actual m³ loaded.

Non-conformance: reject, replace, never ship

A piece that fails any check is replaced, not shipped, and the reason is recorded.

- **Polish defect:** reworked on the polishing line, then re-inspected.
- **Size or thickness outside tolerance, chip, crack, vein or colour defect:** rejected. Marked with a red "X" on the sawn bottom face and moved to the reject area, never back into production.
- **Replacement:** cut from the same reserved block where possible. If the block can't supply it, replacement volume is selected under SOP-01.
- **Repeated defects from one block:** stop cutting that block and follow Stage 2, step 5.
- **Record:** every rejection is logged with piece number, reason and photo. Rejected pieces are never used for any StoneCrest export order.
- **Delay:** if replacement affects the loading date, the Order Owner informs the buyer the same day. The standard is not lowered to keep the date.

Records and sign-off

Each order has a complete record from block receipt to sealed container. Records are kept for at least 3 years and can be shown to the buyer or their inspector.

Record	Content	Signed by
Block receipt log	Block numbers received, arrival photos, transit damage	Production Supervisor, QC Inspector
Approved cutting plan	Sizes and quantities per block	QC Inspector
Production log	Random checks at Stages 2 and 3, results, photos	QC Inspector
Final Inspection Checklist	Every piece: measurements, Pass or Reject	QC Inspector
Rejection log	Piece number, reason, photo, replacement	QC Inspector
Third-party inspection report	Where the buyer chose it	Buyer's inspector
Packing list and load photos	Actual pieces, m ³ , weights; loading and seal photos	QC Inspector, Order Owner

Container release: the Order Owner signs the release only when every record above is complete.

Final Inspection Checklist (form)

One row per piece, filled in at final inspection. Thickness and size are recorded in mm.

Field	Entry
Order number	
Size inspected (cm)	
Specified thickness (mm)	
Date	

Piece no.	Block no.	Thickness: 4 corners + centre (mm)	Length x width (mm)	Diagonals equal (Y/N)	Colour, veins, cracks OK (Y/N)	Polish and edges OK (Y/N)	Pass / Reject	Reason if rejected

Pieces passed: _____ Pieces rejected: _____

QC Inspector (StoneCrest): name, signature, date _____

Factory Production Supervisor: name, signature, date _____